Lip Lift

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KEYWORDS

• Facial feminization • Lip lift • Gender-affirming surgery • Gender-confirmation surgery

KEY POINTS

- Gendering-affirming lip lifts must consider known aesthetic ideals of the female upper lip.
- Patient selection, preoperative education, and shared decision-making are key in minimizing complications and optimizing patient satisfaction.
- Patients' preoperative upper incisal show, alar base width, nasal sill prominence, skin tone and thickness, and vertical maxillary height must all be taken into consideration when planning lip lifts.
- The bullhorn-type subnasal lip lift is designed to achieve optimal lift along the entire upper lip (as afforded by alar base width) and hide scars at the nasal base under the nasal sill.

INTRODUCTION

Male and female faces have a multitude of different distinct features that create an overall masculine or feminine appearance. Facial feminization procedures have been designed to address several of these characteristics, primarily targeting the forehead, nose, and jawline.[1] The lips, however, are an aesthetic focal point of the lower face that have gender-specific features. The lip lift shortens the distance from the nasal base to the vermilion border using skin excision and tissue advancement. Although the procedure results in a subtle change, it can dramatically feminize the appearance of the lower face.

The indirect upper lip lift was first described for lip rejuvenation to treat senile upper lip ptosis and vermilion thinning.[2] The subnasal lift, or "bullhorn lip lift," uses a curved excision pattern along the nasal sill and alar creases to remove upper lip skin while hiding the scar along the nasal base, Multiple different variations of this procedure have been described.[3–7] The subnasal lip lift has been demonstrated to significantly decrease total upper lip and prolabium height, and increase vermilion height,[8] creating an upper lip that more closely resembles the described aesthetic ideals.[9]

These changes in upper lip measurements also generally correlate with a shift from masculine to more feminine anthropometric proportions. Lip lifts therefore have become a useful component of facial feminization, and are a powerful tool for changing the gender-associated appearance of the lower facial third. We have adopted a modified "bullhorn" technique to achieve a subtle lift while minimizing visible incisions.

The goals of the lip lift as a gender-confirming procedure are, foremost, to reduce the height of the cutaneous upper lip, increase the visible red vermilion, and enhance pout while attaining an appropriate amount of dental show and minimizing visible scarring. We herein describe the senior author's preferred technique for male-to-female lip lifts in the context of traditional lip rejuvenation procedures and with important considerations for transgender patients.

THE UPPER LIP

Anatomy

The upper lip should be considered with regards to its anatomic and aesthetic subunits. It is a trilaminar structure with the middle foundational layer consisting of the superior half of the sphincteric

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orbicularis oris with contributions from the buccinator and lip elevators. The orbicularis oris is juxtaposed by the mucosa posteriorly containing multiple minor salivary glands. The outermost layer of the lip is divided into the skin and the vermilion. The upper lip contains hair-bearing skin with numerous sweat and sebaceous glands underneath, which is a thick layer of subcutaneous fat. The red vermilion represents a transition from skin (at the white roll or vermilion border) to oral mucosa (at the wet-dry line) and consists of a thin layer of epithelium absent of glandular elements with a rich underlying capillary network. The philtrum, or philtral groove, is the central focal point of the upper lip, defined on each side by philtral columns formed from insertions of the contralateral orbicularis oris with contributions from the levator labii superioris inferiorly.

The aesthetic boundaries of the cutaneous upper lip include the vermilion border inferiorly, the nasal sill and alar bases superiorly, and the nasolabial fold and oral commissures laterally. The upper lip is further divided into aesthetic subunits including two medial subunits comprising the philtral groove and two lateral subunits each formed by a philtral column and lateral lip adjacent to the nasolabial fold.[10] For the purposes of lip lifts, relevant surface anatomy measurements include the subnasale to labiale superius (depth of cupid's bow) in the midline, or philtral distance, and labiale superius to stomion measurement, or labial distance.[11] These measurements correlate to the upper lip cutaneous skin height and vermilion height, respectively, in the midline.

Aesthetic Ideals

Lip lifts performed as gender-confirming procedures must take into account the aesthetic ideals of the upper lip. These ideals include an upper lip cutaneous skin height (philtral distance) to vermilion height (labial distance) ratio of less than three,[11] and a short philtrum with symmetric and prominent philtral columns.[12] Post lip rejuvenation lip lifts have an ideal philtrum to upper vermilion ratio of 2 to 2.9 and visible upper to lower vermilion of 0.75 to 0.8.[11,13] In general, a greater upper vermilion height, or vermilion show, has been found to be more attractive.[14] These ratios should be taken into consideration when planning gender-affirming lip lifts to ensure that results are not only more biologically "female," but also aesthetically pleasing.

The Male and Female Upper Lip

Differences in the appearance of the male and female upper lip are the basis of lip lifts as gender-affirming procedures for transgender females. Overall upper lip height, as measured from supramentale to stomion, is greater in males than females (average 23.6 vs 20.6 mm, respectively).[15] Subsequently, females tend to have a greater ideal upper incisor show in repose compared with males (around 4 vs 2 mm, respectively). The proportion of upper lip cutaneous skin height to upper vermilion height is also greater in males.[16] However, upper vermilion height itself is not different between males and females.[15,17] Along these lines, increased upper vermilion height has been found to be more attractive in males and females.[14]

PREOPERATIVE CONSIDERATIONS

All gender-affirming surgeries should follow the recommendations of the World Professional Association for Transgender Health Standards of Care. As with other gender-affirming procedures, the persistence of gender dysphoria, the patient's capacity to make fully informed decisions and consent to treatment, and the presence of any significant medical or mental health disorders should be assessed and documented.[18] Although a lip lift does not require referrals from mental health providers,[18] ongoing mental health support is an important aspect of comprehensive care during gender-affirming procedures. Aside from a general medical and surgical history, particular attention should be paid to prior facial feminization procedures, especially rhinoplasty or mentoplasty, and any history of orthognathic surgery.

Classification systems described for lip rejuvenation can provide a useful template for analysis.[11] Important factors to consider include philtral height, labial height (vermilion height), and dental show. The position of the oral commissures and lower lip vermilion height and fullness are also taken into consideration. Absolute measurements and relative proportions are important. Surrounding features should be evaluated including lower facial height, chin projection, nasal tip projection, nasolabial angle, and alar base width, at a minimum. An upper lip lift is roughly simulated with bimanual traction to estimate the appropriate amount of resection and gauge a general postoperative result.[3] Any preoperative asymmetry should also be noted and shown to the patient.

Underlying bony anatomy and general cephalometric concepts must be taken into account when planning soft tissue changes. Patients with vertical maxillary excess will have increased upper incisor show in repose and a "long face," whereas those with vertical maxillary deficiency will have an absence of incisal show on smiling. Occlusive

patterns have a similar effect on upper lip height because a more protruded maxilla (sella-nasion-A point angle >82°) may convey the appearance of decreased upper lip height. Changes from manipulation of the soft tissues may therefore be exaggerated or minimized by the underlying bony and should be taken into account.

Proper patient selection and evaluation is critical in optimizing outcomes. Patients that will benefit from a lip lift have an increased philtral height without significant upper incisal show in repose. Patients with a wider alar base will have a more effective lateral lift as the excisions extend to the perialar creases, whereas those with a narrower base width may only achieve adequate central lift. A nasal base to lip-width ratio of 1:2 has been suggested to prevent inadequate lateral lifting.[19] Fair-skinned patients with no history of hypertrophic scarring or keloid formation and thinner skin, and older patients, tend to form less readily noticeable scars. Scarring history is particularly important in lip lifts because poor scarring is readily noticeable. It is critical to discuss the scar preoperatively, especially in young patients, and even draw it out for the patient.

Patient education through a process of shared decision-making is paramount in the preoperative work-up. Standard details on the procedure, recovery, and potential complications should be appropriately communicated and patient comprehension ensured. Important points include expected time to healing, duration of postoperative swelling, and evolution of scar appearance. This process additionally involves listening to the patient's own goals and mutually arriving at realistic expectations of postoperative results for this procedure. Preoperative and postoperative images from prior patients are helpful in this regard.

Although top surgery and genital surgery are routinely covered by insurance, facial feminization procedures are often categorized by insurance providers as "aesthetic." Although these operations are as effective in treating gender dysphoria in certain cases, patients must understand the financial implications of this labeling by insurance providers. Fees and patient costs should be discussed and agreed on preoperatively.

SURGICAL TECHNIQUE

Markings

Preoperative markings are performed with the patient in the sitting position before infiltration of local anesthetic. The superior incision line (final scar line) is marked at the junction of the nasal base and upper lip. This line proceeds from the alar base, along the inside of the nasal sill and across the base of the columella (**Fig. 1**). The superior line is similarly drawn on the contralateral side, ensuring symmetry. Vertical lines at the philtral columns and alar bases are then drawn as for reference. The planned amount of skin excision is measured with calipers at these points based on the desired amount of upper lip height and incisal show. This height is usually from 5 to 10 mm and is approximated with bimanual traction or pinch technique. The inferior incision line is then drawn to parallel the superior line, ensuring symmetry between the two sides unless corrections are necessary to correct baseline asymmetry. Markings are finally rechecked and remeasured with calipers.

Operative Steps

The procedure is performed either under local anesthesia with sedation or general anesthesia. The incision lines and planned resection are infiltrated with 1% lidocaine with 1:200,000 epinephrine and adequate time for vasoconstriction is allowed. The incisions are made sharply and dissection is carried to the orbicularis oris with electrocautery. The skin and subcutaneous tissue of the planned excision are then removed off of the muscle and hemostasis is obtained. Wound closure is achieved with 5–0 Vicryl suture in the deep dermal layer and interrupted 6–0 nylons for the epidermis. Meticulous technique during

Fig. 1. Preoperative markings for gender-affirming lip lift. The planned scar line is outlined along the nasal base, running around the alar bases, nasal sill, and columella. Vertical lines are drawn along the philtral columns and at the lateral edges of the alar bases to ensure symmetry. The planned height of excision is estimated using bimanual traction on the upper lip and the lower incision line is measured based on the desired amount of upper incisal show and final upper lip height.

closure is needed to achieve precise dermal and epidermal approximation and minimize aberrant scarring (**Fig. 2**). The wound is dressed with antibiotic ointment.

Postoperative Care

Patients are instructed to apply antibiotic ointment to the incision line for the first postoperative week. Showering is allowed after 48 hours. Postoperative swelling is to be expected and discussed with patients preoperatively. Head elevation, cool compresses, and avoidance of strenuous activity for the first week postoperatively are encouraged to minimize postoperative edema. Nylon sutures are removed in the office in 5 to 7 days after which small adhesive tape strips are applied for further tissue support. Patients are instructed to avoid sun exposure to the scar line for 1 year postoperatively and use sunscreen over the incision when outside. Secondary scar treatments are used, such as scar creams and silicone sheets, and adjunctive treatments including dermabrasion and laser therapy are considered for dissatisfaction with scarring after 6 months.

DISCUSSION

The goals of gender-affirming lip lifts align well with those of lip rejuvenation. These include decreasing the height of the cutaneous upper lip, and increasing vermilion show and upper lip pout. In gender-affirming cases, focus is placed on shortening the upper lip length to within a range that is among aesthetically pleasing female "norms" because total upper lip height and the ratio of philtral to labial height (cutaneous to vermilion height) are greater in males. Importantly, soft tissue changes should aim to achieve an appropriate amount of upper incisal show and balance with the patient's lower facial aesthetics. As usual, undercorrection is better than overcorrection, and measured changes must be precise and tailored to the individual patient's facial profile. That said, a subtle amount of overcorrection accommodates for eventual upper lip descent over time postoperatively. An upper lip lift provides additional benefits of increase upper vermilion show and upper lip pout, which tend to be more aesthetically desirable in males and females.

A variety of different surgical techniques have been described to augment the lips and shorten the subnasal upper lip skin. The original bullhorn technique was described more than 40 years ago,[2] and has persisted as a common method of lip lifting. Several other types of indirect lip lifts have been designed to better address the philtrum[4,20] or lateral lip subunits,[21] conceal scars,[3] or minimize incisions.[22] The traditional subnasal bullhorn or seagull lip lift and its modifications are often used in feminization of the face.[1,23,24] We similarly use this technique because it allows varying degrees of lift with a concealed scar at the nasal base.

Certain differences between gender-affirming lip lifts and lip rejuvenation lifts should be recognized. Gender-affirming lifts primarily focus on shortening the nasal base-to-vermilion border

Fig. 2. (*A*) Preoperative photograph of a transgender female. Note the long philtral length (nasal sill to vermilion border height) and minimal vermilion show. (*B*) Postoperative photograph at 3 months showing decreased nasal base to vermilion border length and increased, symmetric vermilion height with scars along nasal base.

distance because this has been found to be the significant difference between male and female upper lip anthropometric proportions.[15,16] Although increased vermilion show and lip pout have been found to be aesthetically pleasing, these are of secondary importance in gender-affirming procedures because these traits are similar between males and females.[15,17] Furthermore, transgender females desiring a more feminine lower facial third may not have an increased philtral height amenable to lifting. Performing an excision in these patients may lead to undesirable dental show and "gummy" smile, which is difficult to correct. Overresection in any case can have a similar undesirable outcome. Certain authors have suggested that no more than 25% of the upper lip height from nasal sill to vermilion border should be excised in facial feminization cases.[23]

Patient selection and precise surgical technique are critical in minimizing complications. Scarring is usually the primary concern of the patients, and although is inevitable, is concealed with correct incision placement at the interface of the nasal and upper lip aesthetic subunits and with meticulous surgical technique, including gentle tissue handling and precise dermal apposition. Adverse scarring has been reported from 1% to 5% of indirect lip cases in patient series.[3,4] Secondary treatments including dermabrasion, silicone sheets, and laser therapy are used for visible scarring.[25] Older patients tend to heal with less noticeable scars and also have greater preoperative lip ptosis making them better candidates for lip lifts. Patient with darker complexion or history of wide/elevated scars should be counseled preoperatively.

Wound breakdown is uncommon, but is managed with local wound care and secondary scar revision as needed. Other complications include asymmetry, and overcorrection or undercorrection. Although undercorrection can be revised, overcorrection is difficult to treat, and therefore surgeons should err toward a subtler lift, particularly in patients with vertical maxillary excess or borderline incisor show. Alar base distortion is more common with endonasal techniques,[3] but can also be problematic with larger lifts or significant tension at the incision.

Satisfactory aesthetic outcomes similarly rely on selection of appropriate surgical candidates and meticulous preoperative planning. Both the surgeon and the patient must understand what the lip lift can and cannot achieve with regards to feminizing the perioral region. Subnasal lip lifts primarily address the central upper lip and are less effective laterally. This discrepancy is further exaggerated in patients with a narrow alar base width. Postoperatively, decreased lateral lift may manifest as downturned lateral commissures, which usually resolves after a few weeks. Variations in philtral morphology may also affect lip lift results. Central insertion of nonparallel philtral columns has been suggested to result in lateralization of the columns' insertion with lip lift and a more truncated lip appearance.[26] Furthermore, the nasal sill should be taken to consideration because patients with a more prominent sill can better hide incisional scars in the subsill crease.[27]

Lip lifts performed for lip rejuvenation are often combined with lip augmentation. Dermal grafts are performed simultaneously to increase vermilion border fullness.[23] Various lip fillers can also be used to enhance the vermilion and upper lip pout. Because gender-affirming lip lifts are primarily focused on shortening the subcutaneous subnasal lip, or philtral distance, adjunct lip augmentation is performed less frequently as a concomitant procedure. Providers, however, must still be cognizant of patients' history of prior lip fillers. These substances may include non–Food and Drug Administration–approved materials, such as injectable silicone, which can cause significant local inflammatory reactions, granulomas, and infection requiring excision.

Outcomes in facial feminization are continuing to be defined. Particularly with less-commonly performed procedures, such as lip lifts, there is a need for outcomes-based research. Patient-reported outcomes have been reported in facial feminization,[28] but overall, validated measures of these outcomes are lacking for gender-affirming procedures.[29] Moving forward, better assessment of short-term objectives and long-term patient satisfaction will help better define the aesthetic ideals for gender-affirming lip lifts and the optimal techniques to achieve these outcomes.

SUMMARY

Lip lifts are a powerful tool for facial feminization that allow for shortening of the height of the cutaneous upper lip skin to create a more feminine lower facial-third. Lip lifts also provide more aesthetically pleasing increased vermilion height and lip pout. Traditional subnasal bullhorn lifts work well for facial feminization in appropriately selected patients. There are several important characteristics to consider in evaluation of patients including skin type, soft tissue characteristics of the lip and nose, and underlying bone structure that can affect the efficacy of the lip. Proper patient selection, precise preoperative planning to avoid overcorrection and asymmetry, and meticulous surgical technique to minimize visible scarring optimize outcomes.

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